Replication of "Errors Are Aversive"
by Greg Hajcak & Dan Foti
(2008, *Psychological Science*)

Melissa Lewis and Michael Pitts
mlewis@alumni.reed.edu
mpitts@reed.edu

1 Introduction

The error-related negativity (ERN) is a negative-going event-related potential (ERP) component that peaks very quickly after a commission error, and owing to its modulation by affective state and its maximal location being over the anterior cingulate cortex, could be functionally related to defensive motivational states. That is, it may have some relationship to physiological responses associated with the organism being in danger. The acoustic startle response is a well-known physiological marker of defensive state and is measured via electromyography (EMG) at the orbicularis muscle just below the eye. If the ERN is indeed related to defensive motivation, its magnitude is likely to correlate with the magnitude of the startle response.

The original experiment had participants perform a flanker task, a simple cognitive task in which one must indicate whether a central arrow, surrounded by additional ("flanking") arrows, is pointing to the left or right. Response errors are more common for trials in which the direction of the flanking arrows is incongruent with the direction of the central arrow, allowing measurement of the ERN. Additionally, a brief 105 dB auditory "noise burst" stimulus was presented following both trials in which the participants made an error and when they responded correctly, to allow the measurement of the acoustic startle response with EMG. The experiment demonstrated a robust ERN, as well as an "error-potentiated startle" (EPS) response (larger EMG signal elicited by noise bursts that followed errors relative to correct responses). The novel finding was that the magnitude of the EPS was significantly correlated with the amplitude of the ERN.

2 Methods

2.1 Power Analysis

The main finding of the original study was a correlation between ERN and EPS amplitudes, r = -.38, $p < .05$, one-tailed. In order to achieve a power of .8, we planned to collect data from 42 subjects.

2.2 Planned Sample

The sample should be adult human subjects 18-35 years old, with normal or corrected-to-normal vision and no history of neurological disorders.

2.3 Materials

Table 1: Methods of the original and replication studies.

Hacjak & Foti 2008	**Replication**
"Continuous electroencephalographic (EEG) and electromyographic (EMG) activity was recorded using an ActiveTwo head cap and the ActiveTwo BioSemi system (BioSemi, Amsterdam, The Netherlands)."	We used a custom EasyCap electrode cap and a BrainAmps Standard EEG system (Brain Products, Gilding, Germany).
"Recordings were taken from 64 scalp electrodes based on the ten-twenty system, as well as from two electrodes placed on the left and right mastoids. The electrooculogram (EOG) generated from blinks and eye movements was recorded from four facial electrodes: two approximately 1 cm above and below the participant's right eye, one approximately 1 cm to the left of the left eye, and one approximately 1 cm to the right of the right eye."	Our electrode caps had 96 scalp electrodes at equidistant positions including two electrodes placed on the left and right mastoids. EOG was recorded identically to the original study.
"The startle response was measured with two electrodes placed approximately 12 mm apart under the participant's left eye on the obicularis muscle. As per BioSemi's design, the ground electrode during acquisition was formed by the Common Mode Sense active electrode and the Driven Right Leg passive electrode. All bioelectric signals were digitized on a laboratory microcomputer using ActiView software (BioSemi). Sampling was at 1024 Hz."	As in the original study, EMG electrodes were placed under the eye and 12 mm apart. Though the original article did not describe their placement except in reference to each other, the medial electrode was placed under the pupil and the lateral electrode was placed 12 mm to the right of it, as per Hajcak's directions in an email correspondence. His suggestion about vertical placement of the electrodes was to feel the orbicularis muscle for where the blink is maximal. All bioelectric signals were digitized on a laboratory microcomputer using Recorder software (Brain Products). Sampling was at 1000 Hz.

2.4 Procedure

Hajcak & Foti, 2008

"The task was an arrow version of the flankers task (cf. Hajcak et al., 2005). On each trial, five horizontally aligned arrowheads were presented, and participants had to respond to the direction of the central arrowhead by pressing the left or right mouse button. On compatible trials, all five arrowheads pointed in the same direction (either left or right), and on incompatible trials, the central arrowhead pointed in the direction opposite the direction of the flanking arrowheads. Compatible and incompatible trials were equally frequent, and all stimuli were presented for 200 ms with an intertrial interval that varied randomly from 500 to 1,000 ms. Participants performed eight blocks of 30 trials. At the end of each block, participants received performance feedback designed to encourage fast and accurate responding. If performance was 75% correct or lower, the message 'Please try to be more accurate' was displayed; performance above 90% correct was followed by 'Please try to respond faster'; if performance was between these levels, the message 'You're doing a great job' was displayed. All participants performed one practice block of 30 trials.'"

Replication

Because we were unable to obtain the Presentation code from the original authors, we made our own stimulus presentation program using the same software. When we emailed Dr. Hajcak a copy of our program he agreed it was sufficiently identical to the original.

2.5 Analysis Plan

Analysis will be identical to that of the original study. As with the original study, offline analysis will be performed using Brain Vision Analyzer software (Brain Products, Gilching, Germany). Data will be referenced to the numeric mean of the mastoids and band-pass filtered with cutoffs of 0.1 and 30 Hz. EEG will be segmented for each trial, beginning 200 ms before response and continuing for 800 ms. It will be corrected for blinks and eye movements using EMCP. The ERN will be defined as the average activity in a 0- to 100-ms window following response onset on error trials; it generally peaks approximately 50 ms following commission of an error. The ERN will be statistically evaluated with Greenhouse-Geisser correction applied to p values associated with multiple degrees of freedom, repeated measures comparisons.

Startle response EMG data will be band-pass filtered (28-512 Hz; 24 dB/ octave roll-off), rectified, then low-pass filtered at 30 Hz (24 dB/ octave) and baseline-corrected. Response magnitude and latencies will be quantified according to a peak in the 20- to 120-ms window after the startle probe was presented. It will be statistically evaluated with Greenhouse-Geisser correction applied to p values associated with multiple degrees of freedom, repeated measures comparisons.

2.6 Differences from Original Study

Differences in materials were outlined in the above materials section. The original article did not specify an age range, but the age range for this study was 18-35. The original author analyzed data using SPSS, while this data was analyzed with R and Python.

3 Methods Addendum

3.1 Actual Sample

Sample was 51 subjects between the ages of 18 and 35, with normal or corrected-to-normal vision and no history of neurological disorders. Eight subjects were excluded from final analysis because impedances were too high (>5kΩ) in at least one of the electrodes in which the statistical analyses were to be performed, or because of excessive muscle artifacts. Thus, the final sample included in the analysis was 43 subjects.

3.2 Differences from pre-data collection methods plan

The original article described the auditory stimulus as being 105 dB, whereas in our replication the intensity was slightly lower because our speakers maxed-out at 99 dB.

4 Results

4.1 Data preparation & analysis

Preparation followed the pre-data collection plan. Our software for ERP and EMG preprocessing (e.g. referencing, filtering, averaging) matched that of the original study. Analysis followed that of the plan preceding data collection, except that we conducted analysis with R and Python rather than SPSS, using the Pearson's product-moment correlation.

4.2 Statistical results

A robust ERN was observed (mean amplitude = 6.44μV; SEM = .73) which was similar in magnitude to the original study (see Figure 1). This difference between ERP amplitudes following errors vs. correct responses was significant ($t(40) = 8.69$, $p < .01$).

An error potentiated startle response was also apparent (see Figure 2), the magnitude of which corresponded closely with the original study (mean difference between error and correct trials = 2.44μV; SEM = 1.17). This effect was significant. Paired *t* tests found that startle magnitude was greater following errors than following either subtype of correct trial, whether predictable, $t(40) = 3.51$, $p < .01$, or unpredictable $t(28) = 2.08$, $p < .05$.

The main finding of the original study, a correlation between ERN and EPS amplitude, did not reach the threshold for significance in our replication: r = -.25, p = .11, one-tailed. The original study reported a significant correlation: r = -.38, $p < .05$, one-tailed. Note that the non-significant correlation in our replication was in the same

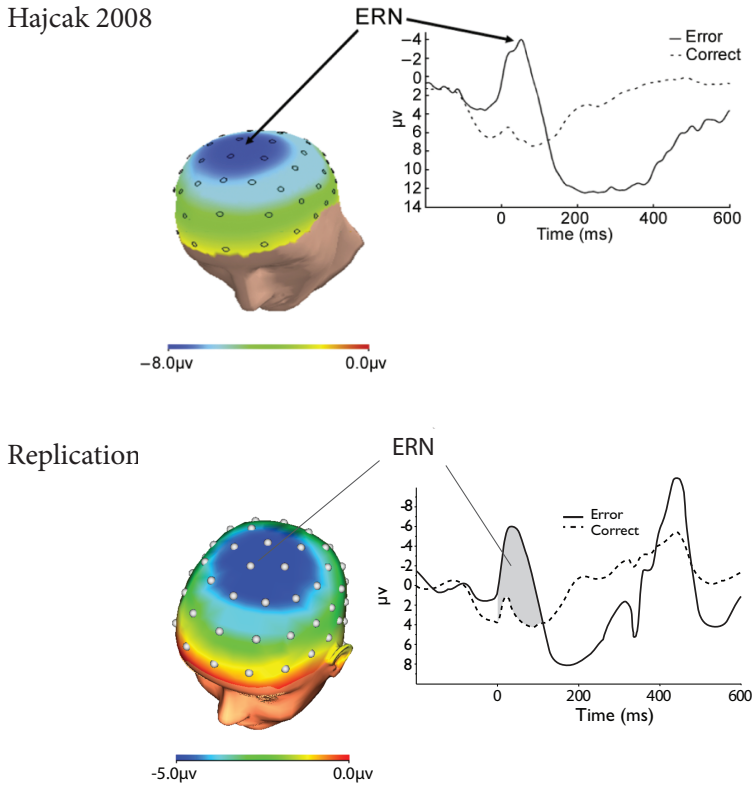


Figure 1: Comparison of mean event-related potentials time-locked to correct and erroneous responses at analogous electrode sites between the original (top) and replication studies (bottom). ERPs (right) plot voltage over time, where response occurs at 0 ms. The ERN's amplitude is calculated as the voltage difference between errors and correct responses in a 100 ms time window. The ERN is plotted over the scalp (left) and is maximal at the frontal-central midline.

direction as in the original study, and the overall pattern of the data was remarkably similar (see Figure 3). Interestingly, both the original study and our replication appeared to show a single outlier subject (in terms of both the ERN and EPS magnitudes being near or greater than 3 standard deviations from the mean) that may have been driving the correlation (see discussion below).

5 Discussion

5.1 Summary of Replication Attempt

While we were clearly able to replicate the basic ERN and EPS results, the main result of the original study was the correlation between the two. The correlation in our replication

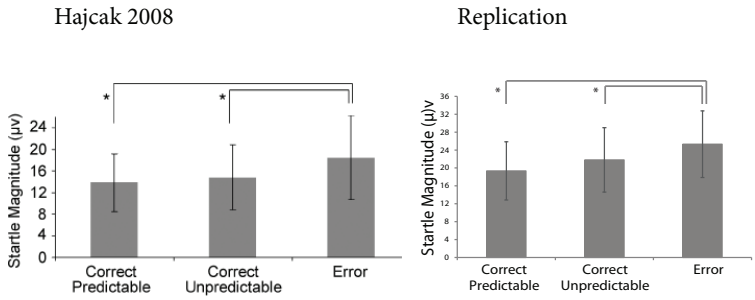


Figure 2: Comparison of mean startle magnitude by trial type from original and replication study. Error bars represent the 95% confident interval. Asterisks indicate significant differences between trial type, $p < .01$

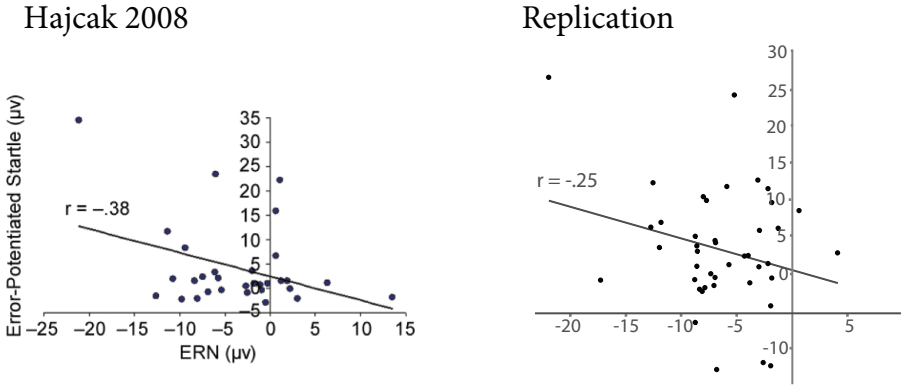


Figure 3: Left: original scatterplot depicting the magnitude of the error-potentiated startle as a function of ERN amplitude. Right: replication findings, not significant ($p > .05$)

trended in the same direction as that of the original study, but did not reach the $p < .05$ threshold for significance. For this reason, we consider this to be a failure to replicate but with extremely similar results.

5.2 Additional Analyses

The main goal of the replication project was to attempt to reproduce a published finding using as close to identical methods and analyses as possible. As noted above, we found extremely similar (albeit non-significant, trending) effects as were reported in the original article. However, a different (and more challenging) goal is to determine whether a reported effect is genuine. In other words, assessments of reliability are different than assessments of validity. The current results, taken at face value, appear to show a correlation between ERN and EPS magnitudes which may have important theoretical and clinical implications. We were interested in exploring the finding of a single outlier subject in both the original and replication studies to determine the impact of these potential outliers on the significant and trending correlations, respectively.

Importantly, one of the original authors attempted to replicate this correlation result in a subsequent study (Riesel, Weinberg, Moran, & Hajcak, 2013, *Journal of Psychophysiology*). They found that the correlation between the ERN and EPS was only significant when assessed in a subgroup of participants who showed large ERNs (via a median split based on ERN amplitude). Therefore, we attempted this same analysis strategy and performed a median split based on ERN amplitude and re-assessed the correlation between ERN and EPS magnitude in this subgroup of participants (see Figure 4). Consistent with the Riesel et al. (2013) study, we found a significant correlation between ERN and EPS in this subgroup of participants (r = -.63, $p < .01$). However, we were concerned that the double outlier, one subject with abnormally large ERN and EPS magnitudes was solely responsible for driving this significant correlation. When we removed the outlier, the correlation dropped to r = -.20, $p = <.4$ (see Figure 5).

In collaboration with one of the original authors, Dr. Hajcak, we performed a meta-analysis across the two data sets (a combined N of 74), and found a significant correlation between ERN and EPS magnitudes, r = -.29, $p < .05$ (see Figure 6). However, once again, when we removed the outliers (now 2 subjects), the correlation was no longer significant (r = -.03, $p = .81$). Because there is no gold standard for identifying outliers (we used the common method of > 2 standard deviations from the mean), it is unclear whether the main correlation reported in the original paper is genuine or not. It could be the case that the correlation exists but is small (being strong in 1 out of every 40 subjects), or it could be the case that in any sample of 40 subjects, noise in the data will cause at least one subject to show abnormally large ERN and EPS magnitudes, thereby strongly influencing the statistical relationship between these two variables.

5.3 Commentary

The greatest impediment to replication was in interpreting the methods from the original article (which took the form of a typical brief research report in the journal *Psychological Science*). Email correspondence with Dr. Hajcak was helpful in clarifying the details of the methods and analyses.

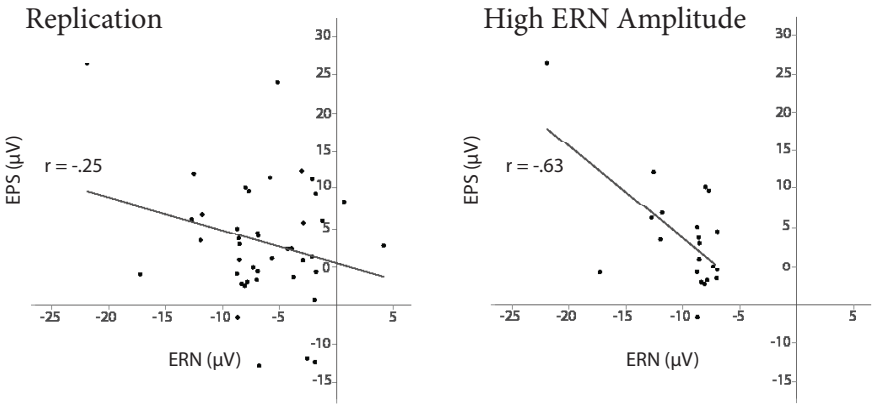


Figure 4: Left: Replication scatterplot (identical to right in 3). Right: correlation plot ($p < .01$) including only participants whose ERN amplitudes were above median (6.91 μV), as performed in original author's replication.

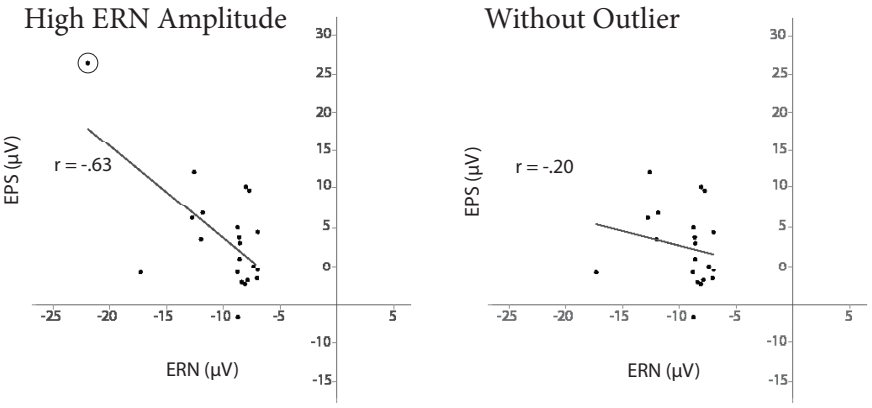


Figure 5: Left: Scatterplot of participants whose ERN amplitudes are above median (6.91 μV), identical to right in Figure 4, outlier (EPS: 2.91σ; ERN: 3.27σ) circled. Right: Same as left, but without outlier ($p = .40$).

It is also unclear to what extent the 6 dB difference between the auditory stimuli used in the original study and the replication may have impacted startle response; however, we do not think this difference contributed to the results because the EPS is a relative dif-

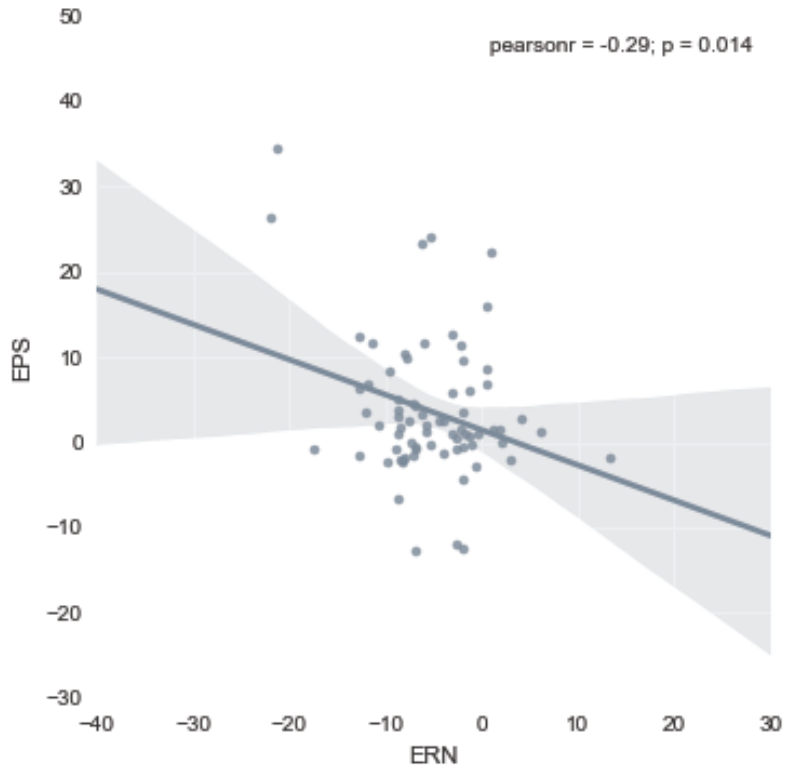


Figure 6: Correlation of ERN amplitude and error-potentiated startle magnitude of combined original (Hajcak & Foti, 2008) and replication participants. Note that if the two outliers on the top left of this figure are removed, the correlation is no longer significant (r = -.03, p = .81).

ference between startle response magnitudes following error vs. correct response trials, and we observed a very similar and significant EPS in our replication.

The identical ERP/EMG analysis software available to the original authors and to ourselves allowed us to precisely follow their methods for data preprocessing. The only technical obstacle was in the original authors being unable to provide the code they wrote for stimulus presentation, though Dr. Hajcak approved of the program written for this replication after we emailed him the program.

The replication experiment revealed a sharp positive ERP peak at ~350 ms followed by a slower negative going peak at ~400 ms, both of which did not exist in the original author?s data (see Figure 1). While the nature of these extra peaks are as yet undeter-

mined, their timing is such that it could they could reflect a post-auricular muscle (or PAM) response to the noise burst, which is well-documented electrical activity evoked in the muscle behind the ear (on the mastoid) following a brief auditory stimulus. It is plausible that this signal would appear in most scalp channels, even one on the top of the head, because they are all referenced to the mastoid electrodes.

We would like to thank the original author Dr. Hajcak for providing input throughout the project. Quality replications are often only possible with close correspondence with the original authors and we appreciate the willingness of Dr. Hajcak to contribute to this replication attempt.